

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2022-23

Zoology

Paper: ZOB-101 — Animal Diversity and Animal Form and Function

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Animal Diversity) (Marks: 35)

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Justify your answer in 2-3 sentences each: [2 × 4 = 8]
- (i) Animals that display radial symmetry are triploblastic.
 - (ii) Development in deuterostomes is marked by radial cleavage and enterocoely.
 - (iii) Cyclostomes are acraniates with paired fins and a bony skeleton.
 - (iv) All chordates are vertebrates but all vertebrates are not chordates.
- (b) Define cloacal respiration. [1]
- (c) Differentiate between neoteny and pedogenesis. [2]

Question 2. Answer the following parts:

- (a) Classify phylum *Cnidaria* **OR** *Echinodermata* up to classes, giving characteristic features and suitable examples of each class. [6]
- (b) Discuss the general characters of *Hemichordata* and enumerate its various classes with suitable examples. [6]

Question 3. Answer the following parts:

- (a) Describe the salient features of class *Amphibia* **OR** *Reptilia* and classify it up to orders, giving characteristic features with suitable examples. [6]
- (b) Explain some of the advantages brought about through the evolution of bilateral symmetry and coelom formation. [6]

Question 4. Write short notes on any two of the following: [6 × 2 = 12]

- (a) Chelicerates
- (b) Cetaceans

- (c) Canal system of Poriferans
- (d) Mosaic cleavage

SECTION – B
(Animal Form and Function) (Marks: 35)

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Justify your answer in 2-3 sentences each: [2 × 4 = 8]

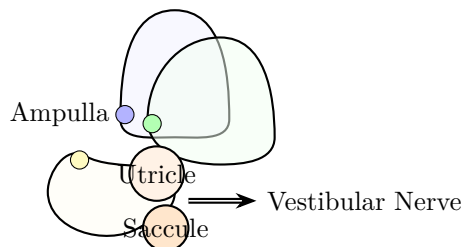
- (i) Apposition image is formed in nocturnal insects.
- (ii) Strobilation is a special type of transverse fission.
- (iii) Amphibians use a negative pressure pump mechanism for ventilating their lungs.
- (iv) Crocodiles have two aortae, the right arising from the left ventricle and the left from the right ventricle.

- (b) Define the following terms: [1 × 3 = 3]

- (i) Ependymal cells
- (ii) Aminotelism
- (iii) External gills

Question 2. Answer the following parts:

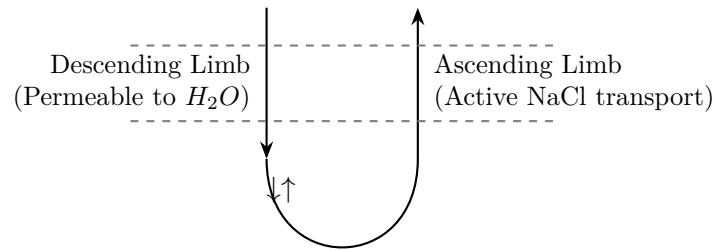
- (a) Describe the role of the vestibular apparatus in the maintenance of body equilibrium in humans, with supportive diagrams. [6]



- (b) With the help of a suitable example, elucidate the food capturing mechanism associated with filter feeding. [6]

Question 3. Answer the following parts:

- (a) Describe the structure and functions of a nephron, giving emphasis on the counter-current exchange mechanism. [6]



- (b) With the help of an appropriate diagram, explain how birds maintain continuous unidirectional airflow through their lungs. [6]

Air Sacs;

ode[draw, rounded corners, thick, fill=orange!10, minimum width=2.2cm, minimum height=1cm, align=center] (lung) at (0,1) Lungs
(Gas Exchange);

ode[draw, rounded corners, thick, fill=red!10, minimum width=2cm, minimum height=1cm, align=center] (ant) at (-3,0) Anterior
Air Sacs;

ode (trachea) at (0,-1.5) [draw, fill=gray!10, rounded corners] Trachea / Outside;

[->, thick, blue, line width=1.2pt] (trachea) to[out=45, in=270] node[right, pos=0.6] Inspiration (post); [->, thick, blue, line width=1.2pt] (post) to[out=135, in=0] (lung); [->, thick, red, line width=1.2pt] (lung) to[out=180, in=45] (ant); [->, thick, red, line width=1.2pt] (ant) to[out=270, in=135] node[left, pos=0.4] Expiration (trachea);